Paleomagnetism of the Lac St-Jean anorthosite and related rocks, Grenville Province, Quebec

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Alternating field and thermal demagnetization study of the Lac St-Jean anorthosite and related rock units in the central portion of the exposed Grenville Province reveals two components of magnetization, one of reversed and the other of normal polarity. Both components are thought to have been acquired during the last regional metamorphism, which was sufficiently intense in this area (mostly amphibolite grade) to reset any earlier magnetization. Corresponding paleopoles at 193°W, 8°S (dm = 7.3°, dp = 4.6°) and 213°W, 19°S (dm = 10.5°, dp = 8.5°) lie along the 950–900 Ma segment of the recently calibrated Grenville track of the North American apparent polar wander path, a track that has thus far been defined largely by results from rock units of the western Grenville.

La désaimantation par champ alternatif et par traitement thermique de l'anorthosite du lac St-Jean et de d'autres unités lithologiques associées à la partie centrale des roches de la province de Grenville révèle deux composantes d'aimantation, une de polarité inverse et l'autre de polarité normale. On croit que les deux composantes furent acquises pendant la dernière phase de métamorphisme régional qui fut suffisamment intense dans cette région (principalement faciès des amphibolites) pour remettre à zéro toute aimantation antérieure. Les paléopôles correspondant à 193°O, 8°S (dm = 7,3°, dp = 4,6°) et 213°O, 19°S (dm = 10,5°, dp = 8,5°) se situent sur le segment 950–900 Ma de la courbe du Grenville récemment étalonnée pour la dérive apparente des pôles de l'Amérique du Nord, une courbe qui jusqu'ici avait été principalement définie par les résultats provenant d'unités lithologiques de la partie ouest du Grenville.

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Introduction

The tectonic history of the Grenville Province has been interpreted, on the basis of paleomagnetic evidence, by several authors (e.g., Fahrig *et al.* 1974; Irving *et al.* 1974; McWilliams and Dunlop 1978; Dunlop *et al.* 1980). With few exceptions, however, paleomagnetic sampling has been confined to the western portion of the Grenville Province in Ontario and the Grenville Front region in eastern Quebec and Labrador. Hence, it is difficult to obtain a coherent picture of the paleomagnetism of the province as a whole.

In this study, alternating field (AF) and thermal demagnetization results are reported for the Lac St-Jean anorthosite and surrounding monzonites, granites, and diabase dikes from the central portion of the Grenville Province.

Geology

The Lac St-Jean anorthosite forms part of a chain of similar intrusions that stretches from the Labrador coast to the Adirondacks. This, the largest body, exceeds 20 000 km^2 in area. The central parts consist essentially of flat-lying layers (0.5–2 m thick) of coarse-grained, black, massive anorthosite. The body is increasingly deformed and metamorphosed near its eastern margins where it has been converted into a sugary, white biotite–plagioclase gneiss (Kehlenbeck 1972). Although the Lac St-Jean intrusion is chiefly anorthosite (or metamorphosed equivalents), parts of it consist of norite or troctolite. A contact metamorphic aureole in gneiss bordering the Lac St-Jean intrusion has yielded a Rb/Sr age of 1451 ± 163 Ma (Frith and Doig 1973). This is the best estimate available for the age of the intrusion.

The Lac St-Jean anorthosite is intrusive into quartz diorite gneiss, granitic gneiss, migmatite, and quartzitic paragneiss. It has closely associated monzonitic rocks, some of which may be genetically related to the anorthosite intrusion. The Ste. Marguerite monzonite (Fig. 1) is an apple green, fine- to medium-grained rock containing uralitized pyroxene. The feldspar and quartz exhibit a granular texture and the feldspar commonly has myrmekitic intergrowths. Rb/Sr work suggests, from initial $^{87}Sr/^{86}Sr$ ratios, that this rock was re-equilibrated about 1100 Ma ago (Frith and Doig 1973). Granodiorite in the Anse St. Jean area (site 29) is similar and has been grouped with the Ste. Marguerite monzonite.

The quartz diorite gneiss in the same vicinity (site 28) probably hosts the monzonite–granodiorite rocks.

The Chamouchouane monzonite (Fig. 1) is a fairly massive subcircular body surrounded by charnockitic gneiss (Benoit 1960). The monzonite is intrusive, and a Rb/Sr isochron dates its consolidation at 1481 ± 94 Ma (Frith and Doig 1973). This age is comparable to that suggested for the Lac St-Jean anorthosite to which it may be related in age if not in origin.

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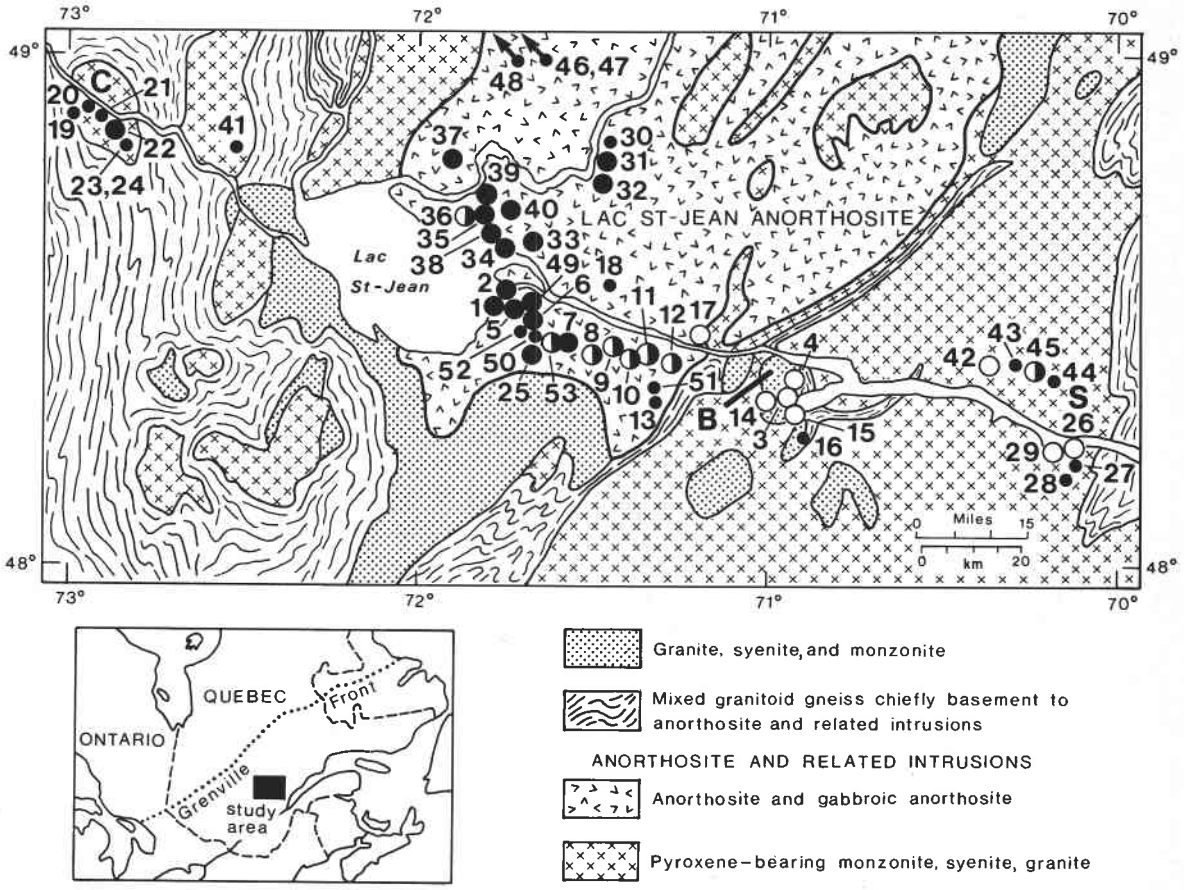


FIG. 1. Geology and sampling sites of the Lac St-Jean area. Sites at which a reversed (normal) magnetization was observed on either AF or thermal demagnetization are shown as large closed (open) circles. Sites of mixed polarity are shown as large half-closed circles, and sites rejected for statistical purposes as small circles. Geology adapted from Laurin (1969). B = Baie des Ha! Ha! granite; C = Chamouchouane monzonite; and S = Ste. Marguerite monzonite.

The Baie des Ha! Ha! body (Fig. 1) is chiefly coarse-grained pink granite containing Carlsbad-twinned potash feldspar megacrysts. It is locally green in colour and the central parts of the intrusion are syenitic. Rb/Sr age determinations of the body are inconclusive, but it was probably partly re-equilibrated about 1100 Ma ago (Frith 1971).

In summary, it seems likely that the main anorthosite intruded older gneissic rocks about 1470 Ma ago. Monzonitic bodies that show a close spacial relationship to the anorthosite were formed at about the same time. Some of the country rock in the vicinity of the anorthosite shows evidence of widespread contact or regional granulite facies metamorphism caused by intrusion and proximity of huge volumes of magma at about 1470 Ma. Long after these magmatic–metamorphic events, the region was exposed to further metamorphic and tectonic disturbances. These effects were not regionally homogeneous and the intensity with which they overprinted the region varies greatly from place to place.

Although the grade of the later "Grenville" metamorphism is known to attain granulite facies (Wynne-Edwards *et al.* 1966), in the Lac St-Jean area this has not been documented and much of the later metamorphism is retrograde, involving hydration of rocks presumably at amphibolite facies grade or less. The timing of these later events is thought to have occurred about 1100 Ma ago.

Sampling and measurement procedures

From 53 sites, 362 cores were collected with a portable diamond drill and oriented with sun and magnetic compasses. Two or more specimens, with diameters of 1 or $1\frac{1}{4}$ in. (2.5 or 3.2 cm) and lengths of $\frac{7}{8}$ or $1\frac{1}{8}$ in. (2.2 or 2.8 cm), were cut from each. The collection included 31 sites from anorthosite, 11 from monzonite, six from granite, two from grano-

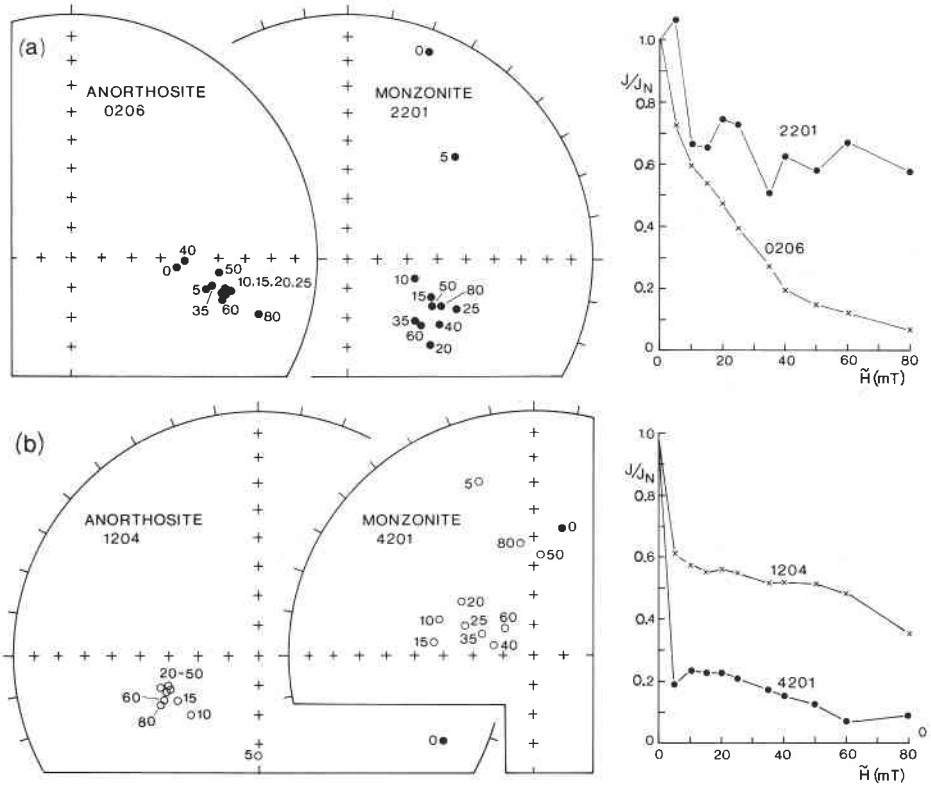


FIG. 2. Typical AF demagnetization directional and normalized intensity results for (*a*) reversely magnetized samples and (*b*) normally magnetized samples. Directions are plotted on an equal-area projection with closed (open) symbols indicating positive (negative) inclinations.

diorite, one from quartz diorite gneiss, and two from diabase dikes (Fig. 1). At anorthosite site 53, samples of a third diabase dike were also collected.

Sample magnetizations were measured on a biastatic magnetometer (Larochelle and Christie 1967) and a commercial spinner magnetometer. Initially two pilot specimens from each site were cleaned by stepwise AF demagnetization to at least 80 mT (800 Oe). Two stable directions of magnetization, one of reversed and the other of normal polarity, were observed. The remainder of the collection was then demagnetized in several (usually two to six) fields selected on the basis of the smallest angular difference between directions of the pilot specimens at a given site. Further specimens were subsequently AF demagnetized in stepwise fashion to investigate the reversal of directions in mixed polarity sites and to ensure that no other stable components were superimposed on those described above.

Twenty-five samples from 19 sites scattered throughout the sampling area and representative of the reversed, normal, and mixed polarities observed on AF demagnetization were thermally demagnetized in stepwise fashion using a commercial thermal demagnetizer. Peak temperatures employed were usually 20, 100, 175, 250, 300, 350, 400, 450, 500, 515, 530, 540, 550, 560, 570, 580, 590, and 600°C. Occasionally temperatures of 475, 555, 565, and 575°C were also included.

AF demagnetization results

Directions of magnetization of samples from 17 anorthosite sites and one monzonite site reach a stable end point with an easterly declination and positive inclination in AF's of 15 mT or less (Fig. 2*a*). This reversed[1] magnetization defines a south paleopole in the Pacific Ocean. Two anorthosite, five granite, and four monzonite sites and 1 diabase dike site exhibit steep negative (i.e., normal) directions (Fig. 2*b*), roughly reversed to the directions observed at reversely magnetized sites. Three additional sites (sites 8, 10, 11), all from the anorthosite, show evidence of mixed (normal and reversed) polarity (Fig. 3). At some "mixed" sites

[1]The definition of reversed and normal magnetizations (and paleopoles) is adopted for convenience. It is not meant to imply that the well defined apparent polar wander path for North America during the last 300 Ma, which extends from the present North pole to the northwest Pacific Ocean, makes a simple connection with the Grenvillian portion of the late Precambrian path.

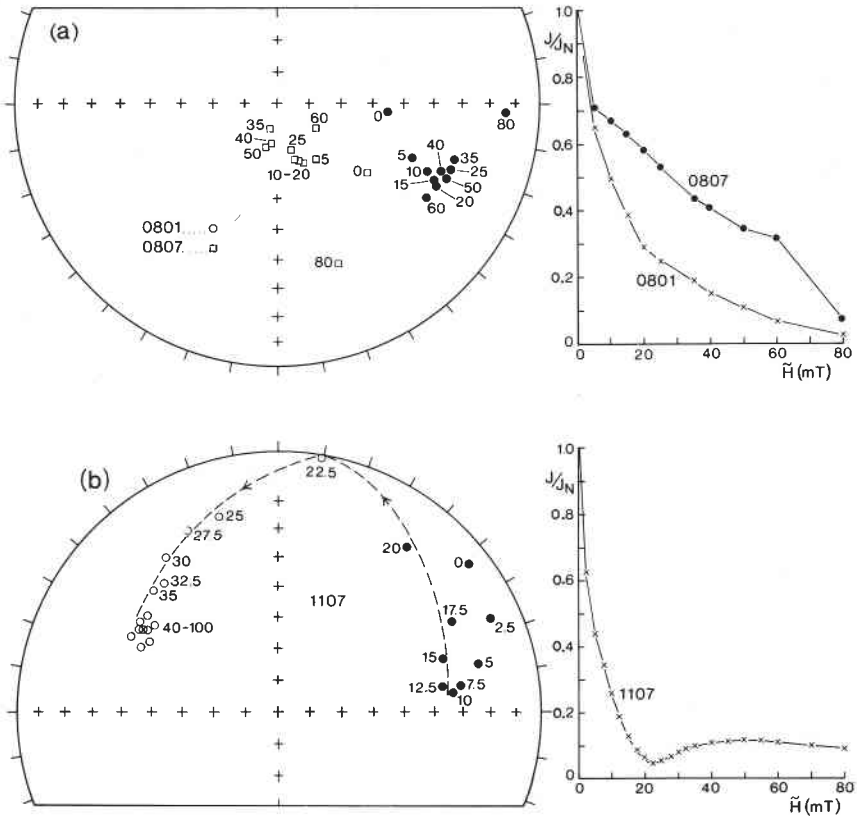


FIG. 3. Typical AF demagnetization results of mixed polarity. Symbols as in Fig. 2.

(e.g., site 8) individual samples maintain stable reversed or normal end points (Fig. 3*a*), whereas at others directions roughly follow great-circle paths between the normal and reversed endpoints. One of the clearest examples of the latter behaviour occurs at site 11 where sample 1107 (Fig. 3*b*) has a great-circle swing between 10 mT (reversed direction) and 40 mT (normal direction). The normal direction is then maintained to 100 mT, the highest peak field applied. Other samples at this site (site 11) show a predominance of either the reversed or the normal direction.

At the remaining 20 sites (largely in monzonite, granodiorite, and gneiss) directions are usually unstable, although in several cases one or two specimens attain the reversed or nomal directions described above. However, results from such sites are not considered in the statistical analysis below. In general, the normal (upward) directions of magnetization are more steeply inclined than are the reversed (downward) directions (e.g., Figs. 2*b*, 3*a*). Whether of reversed or normal polarity, however, the magnetization of anorthosite and granite samples usually exhibits a greater degree of directional stability than that of monzonite samples (e.g., Fig. 2*b*).

Thermal demagnetization results

Sampling sites at which a stable reversed direction was obtained on AF demagnetization exhibit stable reversed magnetizations on thermal demagnetization (e.g., Fig. 4). Blocking temperature ranges are usually

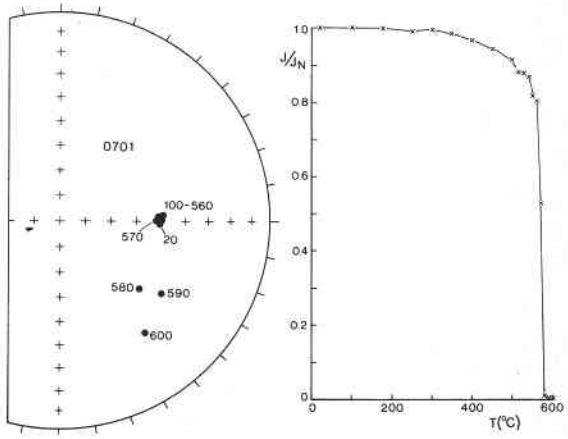


FIG. 4. Typical thermal demagnetization results for reversely magnetized samples. Symbols as in Fig. 2.

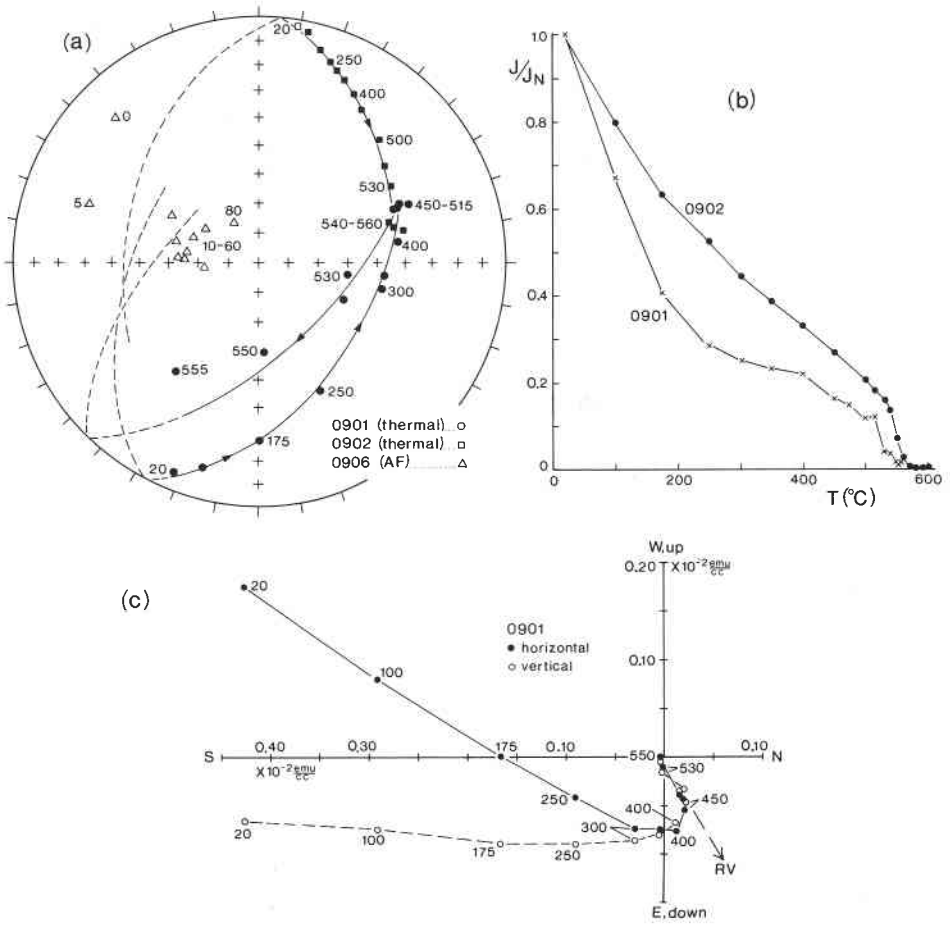


FIG. 5. Thermal demagnetization results for anorthosite site 9. (*a*) Equal-area projection of directional data compared with AF demagnetization results from same site. Symbols as in Fig. 2. (*b*) Normalized intensity data. (*c*) Zijderveld diagram with projection of remanence vectors onto the horizontal (closed symbols) and north–south vertical (open symbols) plane for sample 0901. RV and NV indicate directions of reversed and normal vectors, respectively.

narrow and near 550–580°C, indicating low-titanium titanomagnetite as the dominant remanence carrier.

On the other hand, sampling sites that are normally magnetized on AF demagnetization often show evidence of mixed polarity upon thermal demagnetization (Figs. 5, 6). For example, at anorthosite site 9 (Fig. 5), where samples reach stable normal end points in alternating fields of 10 mT or less, one sample (0902) follows a well defined great-circle path between 20 and 540°C from a shallow northerly inclination to a reversed direction, which is maintained to 560°C. Above 560°C the directions become scattered. A second sample (0901) follows a great circle from a shallow southerly direction and remains stable in the reversed direction between 450 and 515°C before swinging back towards the normal direction at higher temperature. The vector component removed in each temperature interval is analysed in Fig. 5*c* using a Zijderveld plot of the projection of remanence vectors onto horizontal and vertical planes. Although a straight line segment usually indicates that a single paleomagnetic component is being erased during the interval, caution must be exercised. In Fig. 5*c* the straight line between 20 and 300°C (corresponding direction of $D = 210°$, $I = -3°$) undoubtedly indicates elimination of reversed and normal components at the same rate. Beyond 300°C the reversed component increasingly dominates. Between 450 and 550°C only the reversed component is erased, although the swing towards the normal direction above 515°C (Fig. 5*a*) suggests that a small normal component with even higher blocking temperatures is superimposed.

Similarly, at diabase dike site 36, samples display only a normal magnetization upon AF demagnetization, yet, when treated thermally (Fig. 6), both reversed and normal polarities are evident. The direction of magnetization of sample 3607 follows an excellent great-circle arc (Fig. 6*a*) towards the normal end point defined by

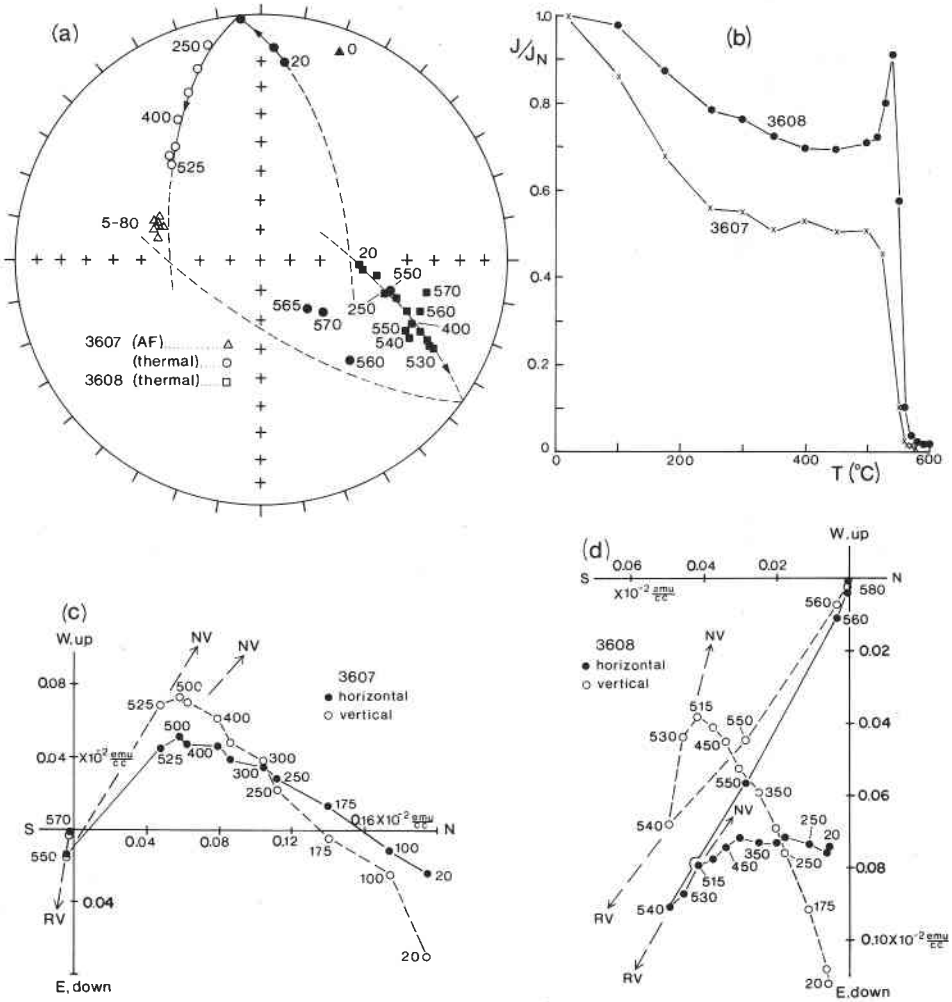


FIG. 6. Thermal demagnetization results for diabase site 36. (*a*) Directional data compared with AF demagnetization results from same site. Symbols as in Fig. 2. (*b*) Normalized intensity data. (*c*), (*d*) Zijderveld diagrams for samples 3607 and 3608, respectively. Symbols as in Fig. 5.

AF results (Fig. 6*a*). However, before the normal end point is attained, and coincident with a drop in intensity from 45 to 10% of its natural remanent magnetization (NRM) value between 525 and 550°C (Fig. 6*b*), the magnetization of sample 3607 shifts to a somewhat scattered but reversed direction. The corresponding Zijderveld plot (Fig. 6*c*) indicates that both components are being eliminated to 525°C, the normal component between 525 and 550°C, and the reversed component above 550°C. The Zijderveld plot of sample 3608 (Fig. 6*d*) from the same site is also complex. To 515°C, both reversed and normal components (or perhaps a viscous component, since the vectors erased lie near the present Earth's magnetic field) are eliminated. In the intervals 515–540 and 540–580°C, normal and reversed components are eliminated, respectively.

Sampling sites that exhibit mixed polarity upon AF demagnetization also show evidence of both reversed and normal components on thermal cleaning. For example, at site 8, individual samples on AF demagnetization are either reversely or normally magnetized (samples 0801 and 0807 of Fig. 7, respectively). However, on thermal demagnetization, the direction of magnetization of sample 0804 follows a short great-circle arc towards, but never reaching, the reversed end point.

The blocking temperature spectra of normal and reversed components at mixed polarity sites are rather variable and often broader than at single polarity sites. As illustrated in Figs. 5–7, blocking temperatures of the reversed component may be either above or below those of the normal component. Since, in some samples (e.g., sample 0804, Fig. 7) neither end point is actually attained, the normal and reversed blocking temperature

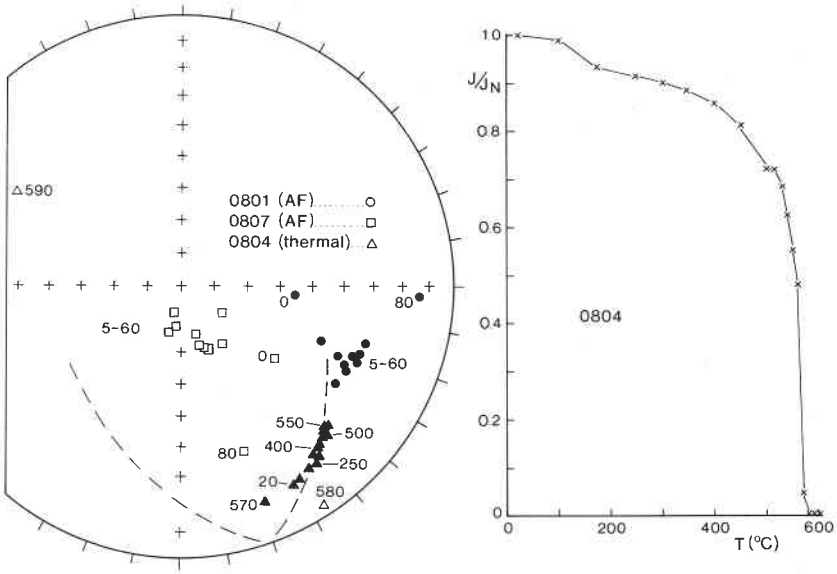


FIG. 7. Comparison of thermal and AF demagnetization of samples from a mixed-polarity site.

ranges may partially or totally overlap. Occasionally (e.g., sample 0901, Fig. 5*a*; sample 3607, Fig. 6*a*) one polarity forms a temperature range part way through that of the other polarity, resulting in a directional swing towards and then away from that end point.

Analysis

Statistical analysis is based on AF demagnetization results. Although thermal work gives essentially the same directional information as AF work, there are insufficient thermal results for statistical purposes. Sites where the number of stable cores, $n \leq 2$, or where normal and reversed magnetizations are difficult to separate are not considered. Mean directions for reversed and normal sites are listed in Tables 1 and 2 and illustrated in Fig. 8*a*.

Anorthosite

Twenty anorthosite sites (117 cores) yield a well defined reversed direction of $D = 109°E$, $I = 45°$ ($k = 32$, $\alpha_{95} = 5.9°$) with a corresponding south paleopole at 192°W, 8°S. Two additional sites (11 cores) have a mean normal direction of $D = 276°E$, $I = -64°$ ($k = 310$, $\alpha_{95} = 14.2°$) and a corresponding north paleopole of 197°W, 29°S. Circles of 95% confidence just fail to intersect, as illustrated in Fig. 8*b*. Although the anorthosite normal direction is based on only two sites, the higher inclination of these normal sites is confirmed by analysis of normal granite and monzonite sites.

Granite

Five sites (27 cores) in granite yield a normal direction of $D = 305°E$, $I = -66°$ ($k = 26$, $\alpha_{95} = 15.2°$), not significantly different from the normal anorthosite direction, but significantly different from the reversed anorthosite direction (Fig. 8*b*). The corresponding paleopole is located at 216°W, 20°S.

Monzonite

Four monzonite sites (19 cores) have a mean normal direction of $D = 312°E$, $I = -67°$ ($k = 257$, $\alpha_{95} = 5.7°$), and a corresponding paleopole at 221°W, 16°S, not significantly different at the 95% confidence level from the normal granite and anorthosite means, but significantly different from the reversed anorthosite mean (Fig. 8*b*).

In addition to the above, one site in monzonite produced a reversed direction and one diabase dike site yielded a normal direction.

Since the normal anorthosite, granite, and monzonite mean directions each differ significantly from the reversed anorthosite direction, overall reversed and normal means have been calculated separately and are given in Tables 1 and 2, respectively. Reversely magnetized sites (20 anorthosite, one monzonite) have a mean direction of $D = 110°E$, $I = 45°$ and a corresponding south paleopole at 193°W, 8°S, and normally magnetized sites (five granite, four monzonite, two anorthosite, and one diabase dike) yield a mean direction of $D = 301°E$, $I = -66°$ and a corresponding north paleopole of 213°W, 19°S. Although the reversed and normal directions are roughly antiparallel, they differ significantly at the 95% confidence level (Fig. 8*c*).

Discussion

Two components of magnetization of opposing polarity are isolated in the Lac St-Jean anorthosite and

TABLE 1. Mean directions and paleopoles[a] for reversed component of Lac St-Jean anorthosite and related rocks

Site	N	n	AF (mT)	D (°)	I (°)	k	α_{95} (°)
				Anorthosite			
1	7	4	25	103	24	130	8.1
2	7	7	25	105	42	156	4.9
5	7	7	35	106	42	119	5.5
6	8	8	50	92	48	90	5.9
7	7	7	35	82	55	161	4.8
8[b]	7	5	40	111	32	213	5.3
10[b]	8	3	25	111	33	234	8.1
11[b]	7	3	15	125	41	99	12.4
25	6	5	25	101	70	28	14.8
31	7	7	20	110	35	51	8.6
32	7	5	25	114	40	62	9.8
33	8	3	25	110	43	157	9.9
34	7	6	60	109	60	41	10.6
35	8	7	60	111	41	151	4.9
37	7	5	40	118	47	193	5.5
38	10	10	60	90	58	108	4.7
39	9	9	60	105	42	257	3.2
40	8	7	60	112	48	97	6.2
49	5	5	30	107	36	134	6.6
53	4	4	30	154	45	234	6.0
All cores		117		107	46	28	2.5
All sites		20		109	45	32	5.9
Mean (south) pole position				192°W	8°S	dm = 7.4°	dp = 4.7°
				Monzonite			
22	6	6	25	122	57	57	9.0
Virtual (south) pole position				210°W	11°S		
				Reversed polarity			
All cores		123		108	46	28	2.5
All sites		21		110	45	32	5.7
Mean (south) pole position:				193°W	8°S	dm = 7.3°	dp = 4.6°

[a]Mean paleopoles are based on unit weight to sites.
[b]Mixed-polarity sites where a reversed direction is isolated in some samples.

associated granite and monzonite (see Fig. 8; Tables 1, 2). In single-component samples, most blocking temperatures are confined to the temperature range 550–580°C, indicating that low-titanium titanomagnetite is the dominant remanence carrier. At mixed-polarity sites, blocking temperature spectra are much more variable, sometimes confined to a narrow range near 580°C, at other times ranging from near room temperature to near 580°C. With blocking temperatures below 580°C both components of magnetization were undoubtedly reset during the last regional metamorphism of the area (mostly amphibolite grade).

Magnetic polarities of the Lac St-Jean anorthosite and associated granite and monzonite show a consistent geographical pattern irrespective of rock type. Sites to the northwest of sites 8 and 18 (see Fig. 1) are reversely (positively) magnetized; those southeast of site 11 (or perhaps 12) are normally (negatively) magnetized; all sites between these limits (i.e., sites 8, 9, 10, 11, 12?) show evidence of mixed polarity upon AF and (or) thermal demagnetization. The only exceptions to this geographical distribution occur in the diabase dikes of sites 36 and 53, which exhibit both polarities, yet are located west of site 8 and mixed monzonite site 45, where thermal demagnetization reveals evidence of both polarities.

The observed difference in inclination between reversed and normal polarities probably occurs thoughout the study area, irrespective of rock type. This, however, is difficult to establish with certainty since most reversely magnetized sites are located in the anorthosite and most normally magnetized sites are found elsewhere. Nevertheless, the directions of two normally magnetized anorthosite sites (sites 9 and 12) are more

TABLE 2. Mean directions and paleopoles[a] for normal component of Lac St-Jean anorthosite and related rocks

Site	N	n	AF (mT)	D (°)	I (°)	k	α_{95} (°)
				Anorthosite			
9	6	5	60	281	−66	255	4.8
12	7	6	60	271	−61	48	9.8
All cores		11		275	−64	72	5.4
All sites		2		276	−64	310	14.2
Mean (north) pole position				197°W	29°S	dm = 22.6°	dp = 18.0°
				Granite			
3	7	5	50	321	−63	48	11.1
4	6	6	50	284	−73	86	7.3
14	6	4	60	318	−44	78	10.5
15	6	6	50	312	−76	116	6.2
17	6	6	50	268	−68	323	3.7
All cores		27		302	−68	27	5.4
All sites		5		305	−66	26	15.2
Mean (north) pole position				216°W	20°S	dm = 25.0°	dp = 20.4°
				Monzonite			
26	8	3	40	305	−66	40	19.7
29	6	3	40	325	−66	105	12.1
42	6	6	25	298	−68	133	5.8
45	7	7	25	320	−67	36	10.2
All cores		19		312	−67	55	4.6
All sites		4		312	−67	257	5.7
Mean (north) pole position:				221°W	16°S	dm = 9.5°	dp = 7.9°
				Diabase dike			
36	8	7	50	295	−55	69	7.3
Virtual (north) pole position:				202°W	12°S		
				Normal polarity			
All cores		64		299	−66	32.9	3.1
All sites		12		301	−66	46.2	6.5
Mean (north) pole position:				213°W	19°S	dm = 10.5°	dp = 8.5°

[a]Mean paleopoles are based on unit weight to sites.

steeply inclined than the directions of most reversed sites (see Fig. 8*a*) and fall among the steep normal directions of granite and monzonite sites.

Perhaps the observed geographical distribution of Lac St-Jean paleomagnetic components may reflect a cooling of the study area through magnetite blocking temperatures (near 580°C) from northwest to southeast or from southeast to northwest, depending on the relative age of the two components. Unfortunately, it is not possible to establish the relative ages of the reversed and normal components from internal paleomagnetic evidence alone. If, for example, both components were acquired as thermoremanent magnetizations (TRM's), one might expect to find higher blocking temperatures associated with the older component. However, in mixed-polarity sites of the Lac St-Jean area, blocking temperatures of one component may be above, below, or overlapping those of the other component (see above). This implies that at least one of the components must be, in part, a chemical remanence.

It is not clear over what time span rocks of the Lac St-Jean area were magnetized. If the time interval was long enough to encompass a reversal of the Earth's magnetic field, but too short for the effects of polar wandering to be observed, then the difference in inclination between reversed and normal magnetizations may best be explained by an asymmetrical field reversal.

If, however, a longer interval of time is involved then the marked difference in inclination between reversed and normal magnetizations likely represents polar wandering. This difference in inclination is evident even in the zone where both polarities occur, suggesting that reversed and normal remanences were acquired at distinctly different times.

The reversal in magnetic polarity occurs in a diffuse

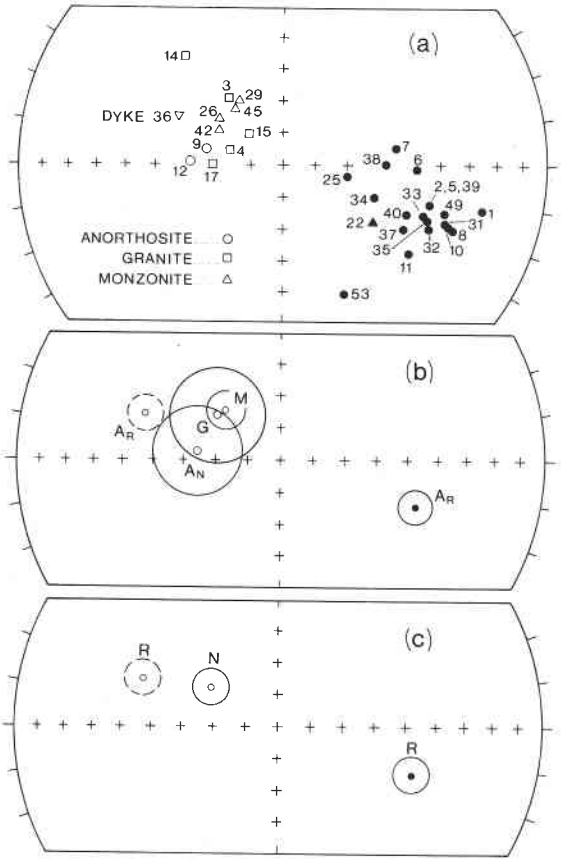


FIG. 8. Magnetization directions after AF cleaning. Symbols as in Fig. 2. (*a*) Site mean directions. (*b*) Mean directions with circles of 95% confidence for reversed and normal anorthosite (A_R and A_N), granite (G), and monzonite (M). The antipole of A_R is shown with a dashed confidence circle. (*c*) Overall reversed (R) and normal (N) mean directions. The antipole of R is shown with a dashed confidence circle.

zone that contains more pristine igneous texture on the northwest side and more deformed gneissic texture on the southeast side. One explanation for the observed pattern of reversal may be the influence of water during retrograde metamorphism. The rocks to the southeast are more hydrous and would retrograde earlier than the drier rocks to the northwest. The maximum temperature attained in the more hydrous rocks would be lower than that in the drier rocks and, in cooling, rocks to the southeast would acquire their normal magnetization before those to the northwest became reversely magnetized.

Grenville paleopoles lie on a broad swath between latitudes 40°S and 25°N in the Pacific Ocean (Fig. 9*a*). In several Grenville studies (Mealy Mountain anorthosite by Fahrig *et al.* 1974; Thanet gabbro by Buchan 1978; Haliburton intrusions by Buchan and Dunlop 1976; Morin anorthosite by Irving *et al.* 1974; White-

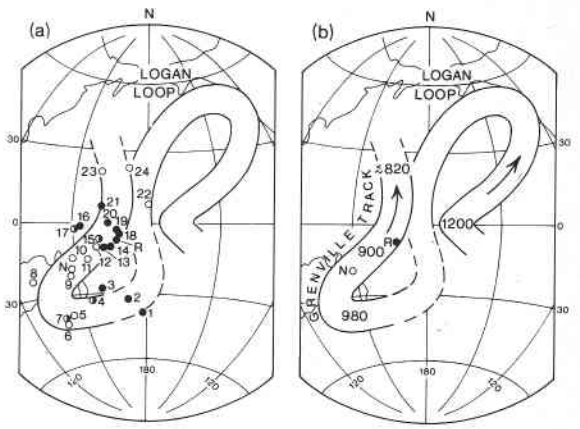


FIG. 9. (*a*) Grenville paleopoles and a portion of the apparent polar wander path for North America. Open (closed) symbols represent north (south) poles. Numbers refer to the poles listed in Table 3. N and R represent the normal and reversed Lac St-Jean paleopoles, respectively. Logan Loop is modified from Pesonen and Halls (1979). (*b*) Lac St-Jean paleopoles relative to the calibrated Grenville apparent polar wander track (see text).

stone intrusions by Ueno *et al.* 1975) two or more paleomagnetic components, strung out along the Grenville "track," have been isolated. Lac St-Jean poles (193°W, 8°S and 213°W, 19°S, respectively) fall roughly midway along the track.

As in the case of the Lac St-Jean magnetizations, the temperatures at which most, if not all, of these Grenville paleomagnetic components were blocked are thought to be less than the peak temperatures attained in these regions in the culmination of Grenville metamorphism. Hence most Grenville remanences were likely acquired in a long cooling period that followed the culmination of metamorphism about 1100–1200 Ma ago (see review in Doig 1977).

Nevertheless, numerous models have been proposed to explain the distribution of Grenville paleopoles and their relationship with those from the rest of North America. They are summarized and discussed by Symons (1978), Roy and W. A. Robertson (1978), Irving (1979), Roy and P. B. Robertson (1979), and Dunlop *et al.* (1980) and can be divided into two types: those that would simply incorporate Grenville paleopoles into a loop in the North American polar wander path (either before or after the 1200–1100 Ma Logan Loop of the North American path) and those that would isolate Grenville poles on a path separate from that for the rest of North America. The latter interpretation is usually invoked on the assumption that Grenville poles, although having different locations from those on the Logan Loop, nevertheless overlap ages or have the same ages as those on the Logan Loop.

Choosing between the above models requires precise

TABLE 3. Grenville paleopoles[a,b]

No.	Rock unit	Paleopole	Reference
1	Mealy Mountain anorthosite (E)	182°W 38°S	Fahrig *et al.* (1974)
2	Thanet gabbro (A_2)	188°W 32°S	Buchan (1978)
3	Thanet gabbro (A_1)	202°W 28°S	Buchan (1978)
4	Ottawa intrusions	205°W 32°S	Irving *et al.* (1972)
5	Haliburton intrusions (Hb_1)	217°W 36°S	Buchan and Dunlop (1976)
6	Morin anorthosite (P)	221°W 42°S	Irving *et al.* (1974)
7	Allard Lake anorthosite	220°W 39°S	Hargraves and Burt (1967)
8	Magnetawan metasediments	230°W 24°S	McWilliams and Dunlop (1978)
9	Whitestone intrusions (WZ)	214°W 22°S	Ueno *et al.* (1975)
10	Wilberforce pyroxenite	212°W 15°S	Palmer and Carmichael (1973)
11	Whitestone intrusions (WW)	204°W 16°S	Ueno *et al.* (1975)
12	Indian Head anorthosite	201°W 10°S	Murthy and Rao (1976)
13	Frontenac Axis dikes	198°W 12°S	Park and Irving (1972)
14	Umfraville gabbro	194°W 11°S	Symons (1978)
15	Umfraville gabbro	200°W 08°S	Palmer *et al.* (1979)
16	St. Urbain anorthosite	206°W 02°S	Robertson and Roy (1979)
17	Grenville dikes	209°W 03°S	Murthy (1971)
18	Whitestone intrusions (WY)	192°W 05°S	Ueno *et al.* (1975)
19	Haliburton intrusions (Hb_2)	193°W 03°S	Buchan and Dunlop (1976)
20	Morin anorthosite (S)	196°W 00°S	Irving *et al.* (1974)
21	Grenville Front anorthosites	199°W 08°N	Palmer and Carmichael (1973)
22	Mealy Mountain anorthosite (NW)	179°W 09°N	Fahrig *et al.* (1974)
23	Thanet gabbro (B)	201°W 20°N	Buchan (1978)
24	Haliburton intrusions (Hb_3)	188°W 25°N	Buchan and Dunlop (1976)

[a]For convenience poles are listed in order of their location on the Grenville Track of Fig. 9*a*.

[b]Two poles (Tudor gabbro, Palmer and Carmichael 1973; Steel Mountain anorthosite, Murthy and Rao 1976), which are believed to significantly postdate the Grenville Track of Fig. 9, are not included.

dating of Grenville paleopoles. In particular, since a long cooling history is likely involved in the Grenville, it is necessary to establish the temperature at which particular magnetic and radiometric systems are "blocked" in order to calibrate the polar wander track (Buchan *et al.* 1977).

$^{40}Ar/^{39}Ar$ incremental-heating studies have recently been used to establish the ages of individual Grenville paleomagnetic components. Using whole rocks Reynolds *et al.* (1978) dated the Frontenac Axis dikes (pole 13, Fig. 9*a*) at ~910 Ma. The Indian Head Complex (pole 12) has been dated at about 880–900 Ma on the basis of $^{40}Ar/^{39}Ar$ incremental-release ages for hornblende (Dallmeyer 1978; Dallmeyer and Sutter 1980). Using $^{40}Ar/^{39}Ar$ incremental-heating experiments on hornblendes, biotites, and plagioclase, Berger *et al.*(1979) have assigned ages of 980 and <820 Ma, respectively, to Haliburton intrusion poles 5 and 24, whose relative ages and approximate blocking temperatures had previously (Buchan and Dunlop 1976) been documented from magnetic evidence. Finally, pole 9 from the Whitestone diorite was dated at ~980 Ma (Dallmeyer and Sutter 1980) from $^{40}Ar/^{39}Ar$ incremental-release ages for hornblende.

A recent attempt to date poles 2, 3, and 23 from the Thanet gabbro (Berger and York 1981) produced $^{40}Ar/^{39}Ar$ incremental-release ages of 1200 Ma (hornblende), 1085 Ma (biotite), and 850 Ma (plagioclase). Unfortunately the relative ages of the Thanet components cannot be inferred from the magnetic data alone (Buchan 1978), and more than one possible interpretation of the ages of these components is possible (see Dunlop *et al.* 1980; Berger and York 1981).

On the basis of the above $^{40}Ar/^{39}Ar$ radiometric data the Grenville polar wander track can be calibrated as shown in Fig. 9*b* (cf. Grenville portion of the North American polar track in Berger and York 1980; Roy and W. A. Robertson 1978). Since most Grenville paleopoles appear to have been acquired between 1000 and 800 Ma ago, the Grenville track is tentatively shown (Fig. 9*b*) to connect with the well defined and somewhat older Logan Loop of the North American apparent polar wander path. Although this is the simplest interpretation of the data, other interpretations of the relationship between Grenville and North American poles are also possible (e.g., Dunlop *et al.* 1980).

As noted above, Lac St-Jean paleopoles fall midway along the Grenville track, at about 950–900 Ma. This age is consistent with magnetization of the Lac St-Jean area during cooling that followed widespread regional

metamorphism. The relationship of the two Lac St-Jean paleopoles with the calibrated Grenville track suggests that the normal remanence is somewhat older than the reversed remanence. Indeed, as noted above, an argument based on the influence of water during retrograde metamorphism supports this age relationship: the more hydrous rocks in the southeast of the study area acquired their normal magnetization before drier rocks to the northwest became reversely magnetized.

Conclusions

(1) Stable directions of magnetization with reversed and normal polarity have been isolated in the Lac St-Jean anorthosite and related rocks. The overall reversed and normal means are $D = 110°$, $I = 45°$ ($k = 32$, $\alpha_{95} = 5.7°$) and $D = 301°$, $I = -66°$ ($k = 46$, $\alpha_{95} = 6.5°$), respectively. Corresponding paleopoles are 193°W, 8°S (dm = 7.3°, dp = 4.6°) and 213°W, 19°S (dm = 10.5°, dp = 8.5°), respectively.

(2) Magnetic polarities of the anorthosite and related rocks show a consistent geographical pattern. Reversely magnetized sites occur in the northwest portion of the study area, normally magnetized sites farther east, and sites of mixed polarity between. Polarities appear not to be related to differences in rock type, since the magnetic reversal is found well within the anorthosite. Likewise, polarities are not correlated directly with magnetic properties such as blocking temperatures since blocking temperatures at single-polarity sites are usually indicative of low-titanium titanomagnetite as the dominant remanence carrier.

(3) The stable magnetizations of the Lac St-Jean anorthosite were undoubtedly acquired after the peak of Grenville metamorphism, since temperatures sufficient to account for the regional metamorphism of the area would unblock any previous magnetization.

(4) Comparison with the calibrated Grenville Track suggests that the Lac St-Jean remanences were both acquired in the interval 950–900 Ma.

(5) Further paleomagnetic study of rock units in the central and eastern portions of the exposed Grenville Province is required before a coherent picture of the tectonic history of the Grenville can be established.

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